Crop Science
Lecture 1

Photosynthesis

Life on earth depend on the energy derived from the sun

Photosynthesis is the only process of biological importance that can harvest energy from the sun

Photosynthesis is the use of solar energy to synthesize carbon compounds

$6\ CO_2 + 6\ H_2O \rightarrow C_6H_{12}O_6 + 6\ O_2$

Energy stored as carbohydrates can be used to power cellular processes in plants and serve as energy source for all forms of life

Light reaction

The active photosynthetic tissue in higher plants is the mesophyll of leaves

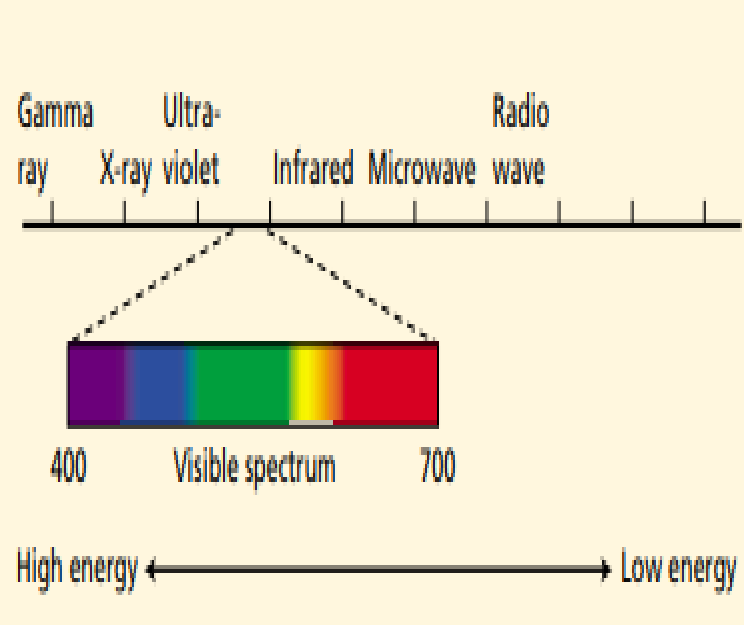
Mesophyll cells contain chloroplasts, the specialized light-absorbing green pigments called chlorophylls

Chloroplast have internal specialized membranes the thylakoids

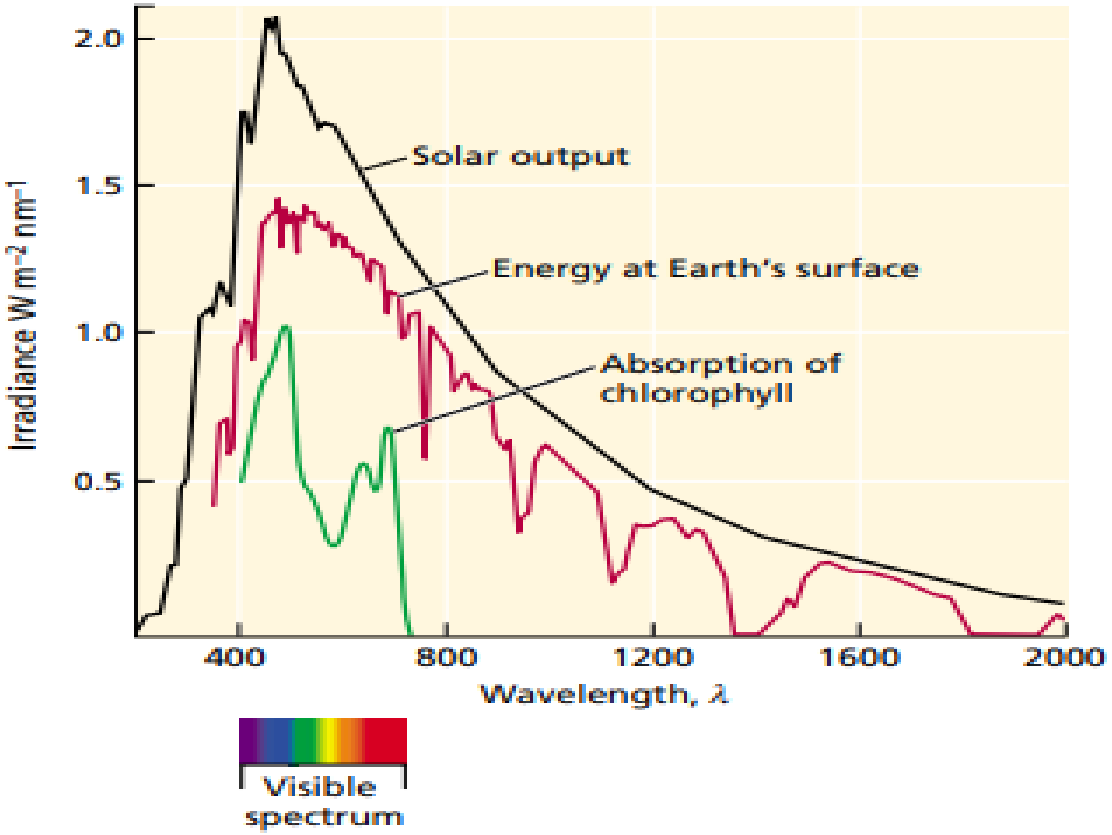
Radiation types

Human eye is sensitive to a narrow range wavelengths of radiation, the visible region, between 400 nm and 700 nm

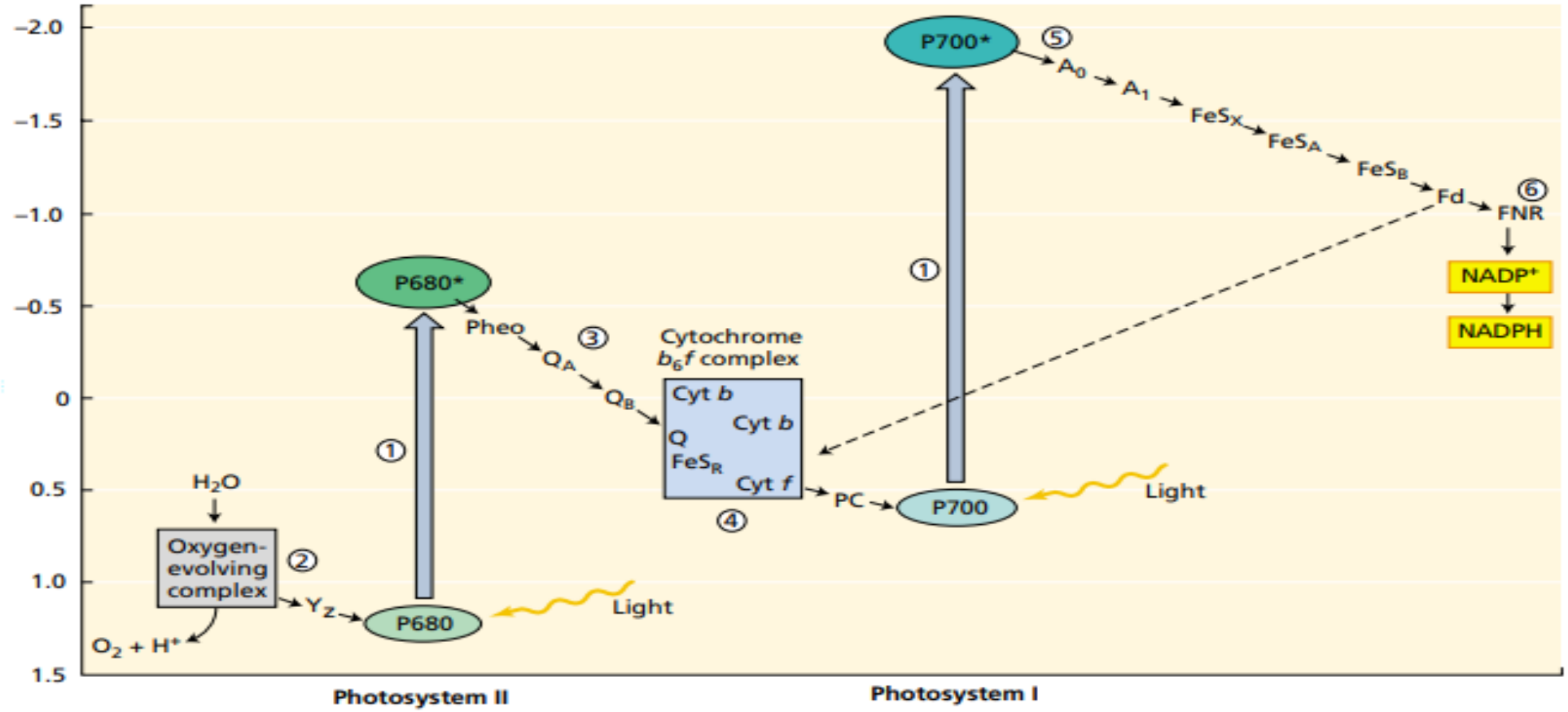
Short-wavelength light has a high energy content and long wavelength light has a low energy content



Radiation types



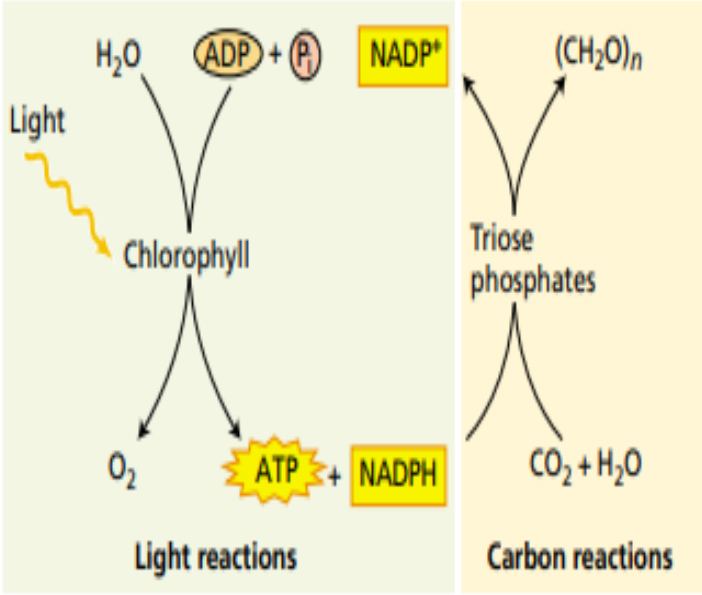
Photosystems



Carbon reactions

Light is required for the generation of ATP and NADPH.

The ATP and NADPH are consumed by the carbon reactions, which reduce CO2 to carbohydrate



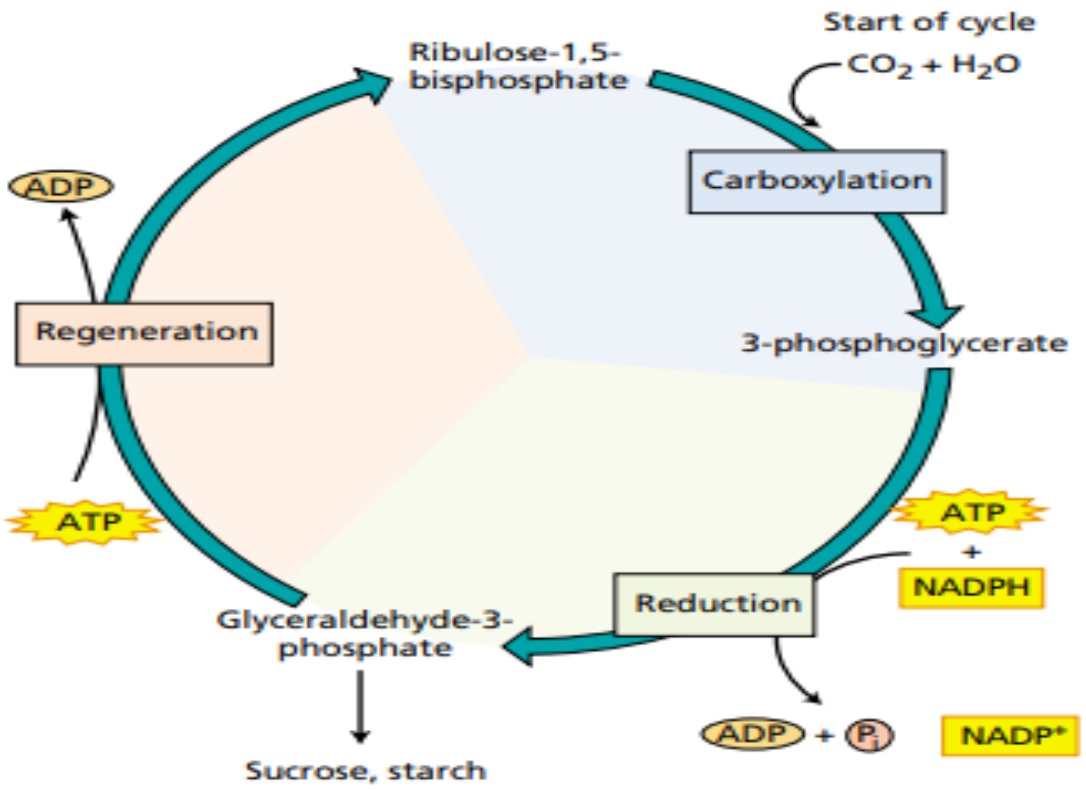
THE CALVIN CYCLE

Plants reduce CO2 to carbohydrate via the photosynthetic carbon reduction cycle called the Calvin cycle

Carbon dioxide and water from the environment are enzymatically combined to form a large chain of carbon molecules

Basic process for Carbon three plant (C3)

Calvin cycle



Calvin cycle application

C4 plants efficiently recycle the carbon back to the cycle

Usually associated with hot environments

Crassulacean plants close stomata during the day and open at night

Associated with desert environment

Physiological and economical considerations

Environmental factors that guide photosynthesis

- Light
- carbon dioxide concentration
- temperature

Leaf structure in response to light or shade environments

- Thin leaf layer in shade
- Shorter palisade cells in shade

Leaf orientation

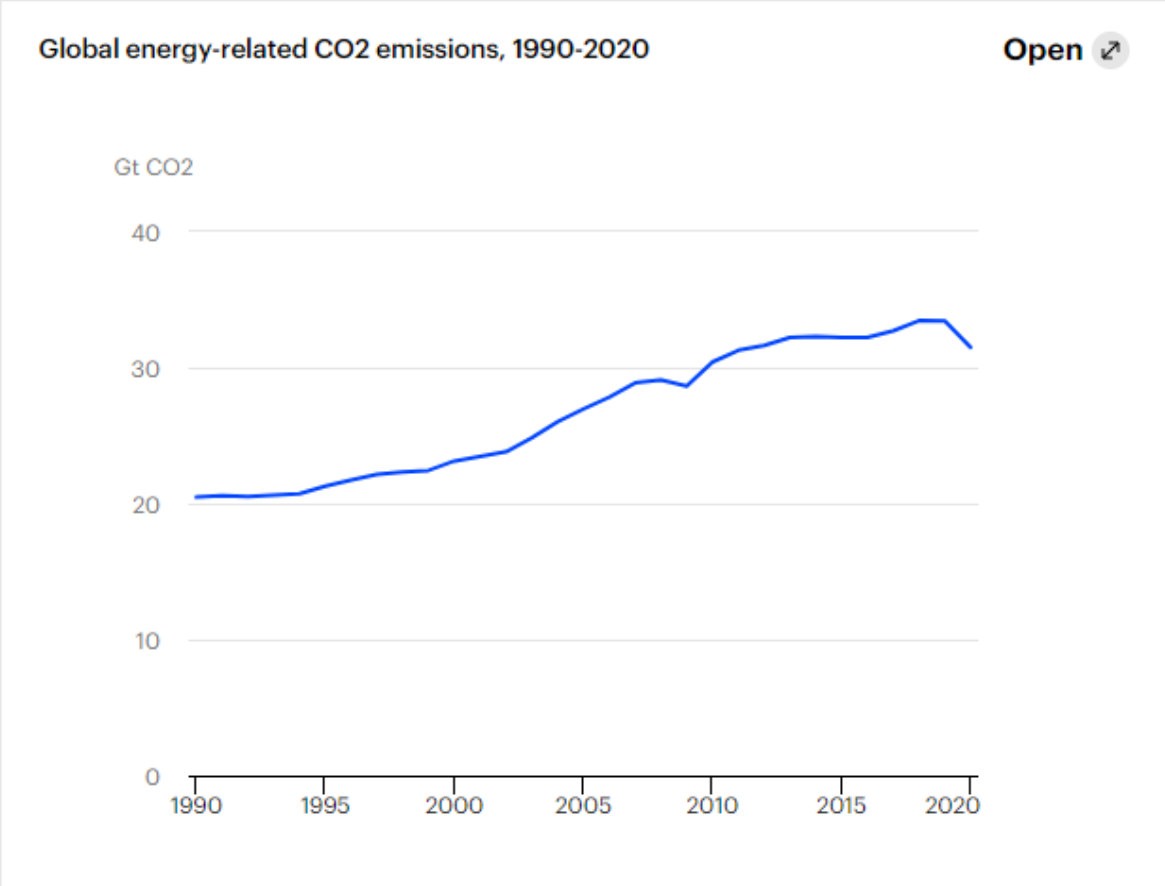
Solar tracking is the leaf orientation toward sun rays

Exposure to sunlight for a long time increases heat and leaf respond by reflecting heat back through

Sensible heat loss – air circulation around the leaf removes heat from surface

Evaporative heat loss – evaporation on the leaf reduce heat energy on leaf

Carbon dioxide emission



Carbon emission

Burning of fossil fuels contributes to increase in carbon dioxide in the atmosphere

Carbon dioxide saturation point of C3 plants is high than in C4 plants